

Codebook ... Job Salary Prediction Dataset

Project: Predicting Job Salaries with Machine Learning **Course:** PSTAT 131/231 Final Project **Dataset file:** job_salary_prediction_dataset.csv **Observations:** 250,000 rows × 10 columns **Missing values:** None (verified in EDA section of report)

Variable definitions

Variable	Type	Description	Possible Values / Range
job_title	Categorical (nominal)	The professional's job title.	AI Engineer, Backend Developer, Business Analyst, Cloud Engineer, Cybersecurity Analyst, Data Analyst, Data Scientist, DevOps Engineer, Frontend Developer, Machine Learning Engineer, Product Manager, Software Engineer (12 levels)
experience_years	Numeric (integer)	Total years of professional experience the individual has accumulated.	0 to 20
education_level	Categorical (ordinal)	The highest education level the individual has attained.	High School, Diploma, Bachelor, Master, PhD (5 levels, ordered)
skills_count	Numeric (integer)	The number of distinct professional / technical skills the individual reports.	1 to 19
industry	Categorical (nominal)	The industry sector in which the individual works.	Consulting, Education, Finance, Government, Healthcare, Manufacturing, Media, Retail, Technology, Telecom (10 levels)
company_size	Categorical (ordinal)	The size of the employer.	Startup, Small, Medium, Large, Enterprise (5 levels, ordered)

Variable	Type	Description	Possible Values / Range
location	Categorical (nominal)	The country in which the role is based, or “Remote” if the role is location-independent.	Australia, Canada, Germany, India, Netherlands, Remote, Singapore, Sweden, UK, USA (10 levels)
remote_work	Categorical (nominal)	Whether the role allows remote work.	No (fully in-office), Hybrid (mixed), Yes (fully remote) (3 levels)
certifications	Numeric (integer)	The number of professional certifications the individual holds.	0 to 5
salary	Numeric (continuous)	OUTCOME VARIABLE. Annual salary in U.S. dollars.	~\$31,800 to ~\$333,000

Notes

- The outcome variable, salary , is right-skewed and is log-transformed for modeling. See the EDA section of the report for the salary distribution plot and skewness diagnostic.
- The dataset has no missing values in any column; this is verified in the report.
- For modeling, numeric features are standardized (mean 0, unit variance) only for the distance- and penalty-based models (KNN, elastic net) and passed through unchanged for the tree-based models (random forest, XGBoost), which are scale-invariant.
- All six categorical features are one-hot encoded, producing 45 indicator columns. Combined with the 3 numeric features this gives 48 model inputs.

Source

Nalisha, A. (n.d.). *Job Salary Prediction Dataset* [Data set]. Kaggle. Retrieved from <https://www.kaggle.com/datasets/nalishaa/job-salary-prediction-dataset> (<https://www.kaggle.com/datasets/nalishaa/job-salary-prediction-dataset>)